Vitamins E and C in the Prevention of Cardiovascular Disease in Men: The Physicians' Health Study II Randomized Trial

Context Basic and observational studies suggest vitamins E or C may reduce risk of cardiovascular disease (CVD). However, few long-term trials have evaluated men at initially low risk of CVD, and no previous trial in men has examined vitamin C alone in the prevention of CVD. Objective To test whether long-term vitamin E or C supplementation decreases risk of major cardiovascular events among men. Design, Setting, and Participants The Physicians' Health Study II (PHS II) is a randomized, double-blind, placebo-controlled factorial trial of vitamins E and C that began in 1997 and continued until its scheduled completion on August 31, 2007. We enrolled 14,641 U.S. male physicians initially aged ≥50 years, including 754 (5.1%) men with prevalent CVD at randomization. Intervention Individual supplements of 400 IU vitamin E every other day and 500 mg vitamin C daily. Main Outcome Measures A composite endpoint of major cardiovascular events (nonfatal myocardial infarction (MI), nonfatal stroke, and CVD death). Results During a mean follow-up of 8.0 years, there were 1,245 confirmed major cardiovascular events. Compared with placebo, vitamin E had no effect on the incidence of major cardiovascular events (both active and placebo vitamin E groups, 10.9 events per 1,000 person-years; hazard ratio [HR], 1.01; 95% confidence interval [CI], 0.90–1.13; P=0.86), as well as total MI (HR, 0.90; 95% CI, 0.75–1.07; P=0.22), total stroke (HR, 1.07; 95% CI, 0.89–1.29; P=0.45), and cardiovascular mortality (HR, 1.07; 95% CI, 0.90–1.29; P=0.43). There was also no significant effect of vitamin C on major cardiovascular events (active and placebo vitamin E groups, 10.8 and 10.9 events per 1,000 person-years, respectively; HR, 0.99; 95% CI, 0.89–1.11; P=0.91), as well as total MI (HR, 1.04; 95% CI, 0.87–1.24; P=0.65), total stroke (HR, 0.89; 95% CI, 0.74–1.07; P=0.21), and cardiovascular mortality (HR, 1.02; 95% CI, 0.85–1.21; P=0.86). Neither vitamin E (HR, 1.07; 95% CI, 0.97–1.18; P=0.15) nor vitamin C (HR, 1.07; 95% CI, 0.97–1.18; P=0.16) had a significant effect on total mortality, but vitamin E was associated with an increased risk of hemorrhagic stroke (HR, 1.74; 95% CI, 1.04–2.91; P=0.036). Conclusions In

this large, long-term trial of male physicians, neither vitamin E nor C supplementation reduced the risk of major cardiovascular events. These data provide no support for the use of these supplements for the prevention of CVD in middle-aged and older men.